

■ INDIAN ARMY ■

Canteen Inventory & Sales Management System

56 APO Field Canteen

A digital solution replacing paper ledgers with automated, role-based inventory tracking, daily sales reporting, and one-click official sign-off.

Version 1.0 · April 2026

Platform: Python · CustomTkinter · SQLite

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01 Problem Statement & Objectives

Current Challenges

- Manual paper ledgers are error-prone and time-consuming for canteen staff
- No automated deduction of raw materials when meals are prepared
- Revenue, COGS, and profit tracked manually — prone to discrepancies
- Stock alerts, shopping lists, and reports prepared by hand daily
- Official sign-off documents require manual compilation each day

Project Objectives

- Replace paper ledgers with a desktop-based digital system
- Automate inventory deduction via a recipe engine
- Provide real-time sales, revenue, and profit visibility
- Generate print-ready, signed daily reports with one click
- Enforce role-based access so officers cannot alter operational data

02 System Overview

A **lightweight, offline desktop application** built in Python with a CustomTkinter GUI and SQLite database. Designed specifically for canteen operations in a low-connectivity military field environment.

Technology Stack

Layer	Technology	Purpose
Frontend (UI)	Python + CustomTkinter	Responsive desktop GUI — big buttons, clear fonts
Backend Logic	Python 3.12	Business logic, recipe engine, report generation
Database	SQLite	Embedded, zero-config, file-based storage
Reports	ReportLab (PDF)	A4 print-ready daily reports with signatures
Deployment	PyInstaller (.exe)	Standalone Windows executable — no Python needed

Deployment

- Packaged as a single **.exe** (Windows) via PyInstaller
- Runs entirely offline — no internet connection required
- Data stored in canteen.db alongside the executable
- Daily automated backup to the backups/ folder

03 User Roles & Access Control

Three clearly separated roles enforce a least-privilege security model:

Role	Default Login	Access Level	Key Responsibilities
Admin	admin / admin123	Full (CRUD)	Master data, user mgmt, DB backup/restore, audit
Manager	manager / manager123	Operational	Sales entry, batch prep, stock mgmt, reports
Officer	officer / officer123	Read-Only	View dashboard, stock levels, daily reports

Permissions Highlights

- **Admin only:** Master Data (items, menus, recipes, users), Manual Audit Adjustments, DB Backup/Restore
- **Manager:** Full operational access — sales, batch prep, stock add/receive, expenditure, reports
- **Officer:** View-only — all write controls are *hidden* from navigation (not just disabled)

Security Policy

- Password-protected login with SHA-256 hashed credentials
- Operators cannot edit historical data without Admin approval
- Navigation items for inaccessible features are fully hidden per role

04 Core Features at a Glance

Module	What It Does
Master Data Management	Item master, menu/product master, pricing setup
Auto-Stock Deduction	Recipe-based batch prep entry auto-reduces inventory
Daily Digital Ledger	Opening → Received → Issued → Closing stock tracking
Sales & Profit Tracker	Revenue, COGS, net daily profit in real time
Smart Inventory Alerts	Colour-coded low-stock warnings + shopping list generator
One-Click Daily Reports	A4 PDF with signature blocks, historical calendar access
Wastage Logging	Records unsold / spoiled food for accurate P&L;
Payment Mode Tracking	Tags revenue as Cash / UPI / Card
User Management	Admin creates & manages user accounts and roles
DB Backup & Restore	One-click backup + restore from any previous archive

05 Auto Stock Deduction — Recipe Engine

The **Recipe Engine** is the heart of the system. It eliminates manual calculation of raw material usage when meals are prepared.

How It Works

- Admin defines a **Recipe** per menu item — exact ingredient quantities for 1 unit
- Manager enters **Batch Preparation Count** (e.g., 150 Lunches prepared today)
- System calculates: *Total Usage = Qty per Unit × Batch Count*
- Inventory is automatically debited — no manual calculation needed
- Wastage can be logged separately to maintain accurate stock records

Example — Standard Lunch (1 Unit)

Ingredient	Unit	Qty / Meal	Usage (150 Meals)
Rice	kg	0.100	15.0 kg
Dal	kg	0.050	7.5 kg
Vegetables	kg	0.080	12.0 kg
Cooking Oil	ltr	0.020	3.0 ltr
Box/Pack	pcs	1	150 pcs

Result: After entering 150 lunches, the system debits the above quantities from inventory automatically, and recalculates COGS in real time.

06 Daily Ledger & Inventory Tracking

Every stock movement is tracked in a structured digital ledger with automated balance calculation.

The Ledger Formula



Key Capabilities

- **Goods Received Entry:** Log new stock purchases with quantity and total cost
- **Category Filtering:** View ledgers for Dry Rations, Fresh Vegetables, Dairy, Packaging
- **Manual Adjustment (Admin only):** Reconcile physical count vs system count with reason logging
- **Reorder Level Alerts:** Set minimum stock limits per item; dashboard shows red warnings
- **Shopping List:** Auto-generate purchase list from all low-stock items

07 Sales, Revenue & Profit Tracking

Real-time financial visibility for every operational day.

Total Revenue	Meals Sold × Selling Price (SP)
Total COGS	Ingredients used × Cost Price (CP)
Net Daily Profit	Revenue – COGS

Additional Features

- **Payment Mode Tagging:** Each sale tagged as Cash, UPI, or Card
- **Wastage Accounting:** Unsold/spoiled meals deducted before profit calculation
- **Daily Dashboard Widget:** Live profit/loss display updated on each entry
- **Historical Data:** Browse and compare profit across any past date

08 Smart Alerts & One-Click Reports

Smart Inventory Alerts

- Configurable reorder levels per item (e.g., Sugar: minimum 2 kg)
- ■ Red dashboard warning badges appear when stock drops below minimum
- Shopping list auto-generated from all items below reorder level

One-Click Daily Report — Contents

Report Section	Description
Opening Balance	Stock at start of day for all items
Purchases (Goods Received)	New stock added during the day
Usage (Issued / Cooked)	Ingredients consumed via recipe engine
Closing Balance	Auto-calculated: Opening + Received – Issued
Meal Sales Summary	Meals sold, wastage, revenue per item
Financial Summary	Total Revenue, COGS, Net Daily Profit
Signature Blocks	Prepared By / Checked By / Approved By JCO/Officer

- Report output is an **A4-optimised PDF** ready to print and sign
- Historical calendar view lets you retrieve any past date's signed report

09 Non-Functional Requirements

Category	Requirement
Usability	Simple enough for non-technical canteen staff with minimal training; large buttons & clear fonts
Accessibility	Mobile/Tablet responsive interface — staff can enter prep numbers from a tablet in the kitchen
Security	Password-protected logins; SHA-256 hashed credentials; operators cannot alter historical data without Admin approval
Data Backup	Daily automated SQLite database backup; one-click manual backup and restore via Admin panel
Performance	Sub-second response for all common operations; lightweight footprint for field-grade hardware
Offline First	No internet required — fully functional in low/no-connectivity military field environments

10 Data Model Summary

The SQLite database contains the following core tables:

Table	Key Fields	Purpose
users	id, username, pw_hash, role, name, rank	Authentication & role management
inventory	id, item, cat, unit, stock, min_lvl, cp	Raw material master & live stock
menu	id, name, sp, active	Finished meal catalogue & pricing
recipes	id, menu_id, inv_id, qty_per_unit	Ingredient–meal linkage (recipe engine)
sales	id, date, meal, sp, sold, wastage, cogs, payment	Daily sales & COGS recording
goods_received	id, date, inv_id, qty, total_cost	Inbound stock purchase log
batch_prep	id, date, menu_id, qty_prepared	Batch cook log → triggers auto-deduction
expenditure	id, date, amount, category, notes	Direct cash expenditure tracking

All tables use **foreign key constraints** (SQLite PRAGMA) and cascade deletes to maintain referential integrity.

11 Sample Daily Operations Report — 06 Apr 2026

Below is a snapshot of an actual report generated by the system for 56 APO Field Canteen.

Meal Sales Summary

Meal Item	Sold	Wastage	Rate	Revenue	Payment
Lunch	10	0	₹10	₹100	Cash
GRAND TOTAL	10	0	—	₹100	

Financial Summary

Total Revenue	Total COGS	Net Daily Profit
₹100	₹30,000	₹ -29,900 (test data)

Inventory Closing Stock

Item	Category	Unit	Opening	Received	Closing	Status
Rice	Dry	kg	10.0	0.0	7.0	OK

Official Sign-Off

Prepared By	Checked By	Approved By
Canteen Manager (JCO)	Supervision Officer (Subedar)	Officer-in-Charge (Captain)
Signature: _____	Signature: _____	Signature: _____
Date: 2026-04-06	Date: 2026-04-06	Date: 2026-04-06

12 Summary & Recommended Next Steps

What Has Been Delivered

- Fully functional **desktop application** — Python + CustomTkinter + SQLite
- 3-tier role-based access (Admin / Manager / Officer) with SHA-256 auth
- Recipe engine for **automated stock deduction** on batch preparation
- Real-time **sales, COGS, and profit** tracking per day
- One-click **A4 PDF daily report** with official signature blocks
- Colour-coded dashboard alerts + auto-generated shopping list
- PyInstaller **standalone .exe** for zero-dependency deployment

Recommended Next Steps

- ■ **Populate master data:** Add all raw materials, menu items and recipes to the system
- ■ **Enter opening stock:** Conduct physical count and enter opening balances
- ■ **Test report format:** Run a sample day, generate and print a report for sign-off approval
- ■ **Schedule backups:** Enable and verify nightly automated database backup
- ■ **Tablet deployment:** Test on tablet for kitchen-floor entry by canteen staff
- ■ **Change default passwords** for Admin, Manager, and Officer accounts before go-live

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